Team :skullpray:

Dakota State University

(Team 39)

Green Team Documentation

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Public Website

Overview

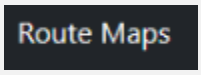
The public website is where you can view bus and train routes and buy tickets to use them. The following describes how to use each of the website's functionalities.

Access the Public Website

1) Open up a web browser, such as Firefox or Chrome. This can be done by pressing the windows key, typing the browser name, and hitting enter or clicking the application.
2) In the address bar, type www.team39.isucdc.com and press enter to access the homepage of the website.

View Bus and Train Routes

1) In the website navigation bar, click "Route Maps".

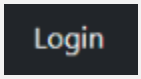


2) Select "Bus" to view bus routes, "Train" to view train routes, or "Both" to see all transit routes and vehicles available.

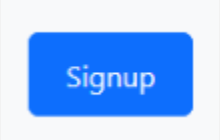
Create a User Account

Note: If you have an already existing account, or have been given credentials to use, please continue down to the "Login to Your Account" section.

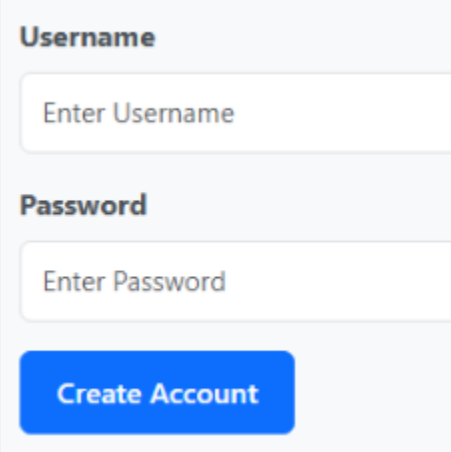
1) In the website navigation bar, click "Login".



2) Scroll down to the bottom of the web page and click "Signup".

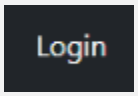


3) Enter your desired username and password in the correct fields, and then click "Create Account".

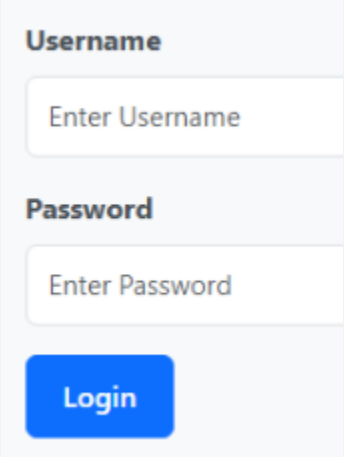


Login to Your Account

1) In the website navigation bar, click on “Login”.

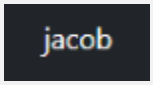


2) Enter your username and password into the correct fields and click “Login”.

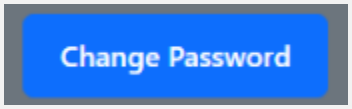


Change Your Password

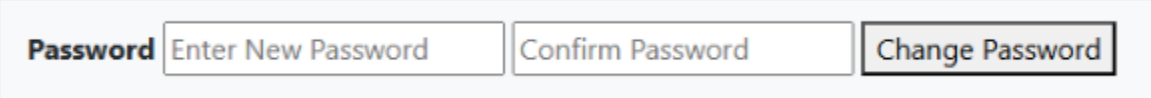
1) In the website navigation bar, click on your username in the top right corner.



2) Under “Account Information”, click “Change Password”.

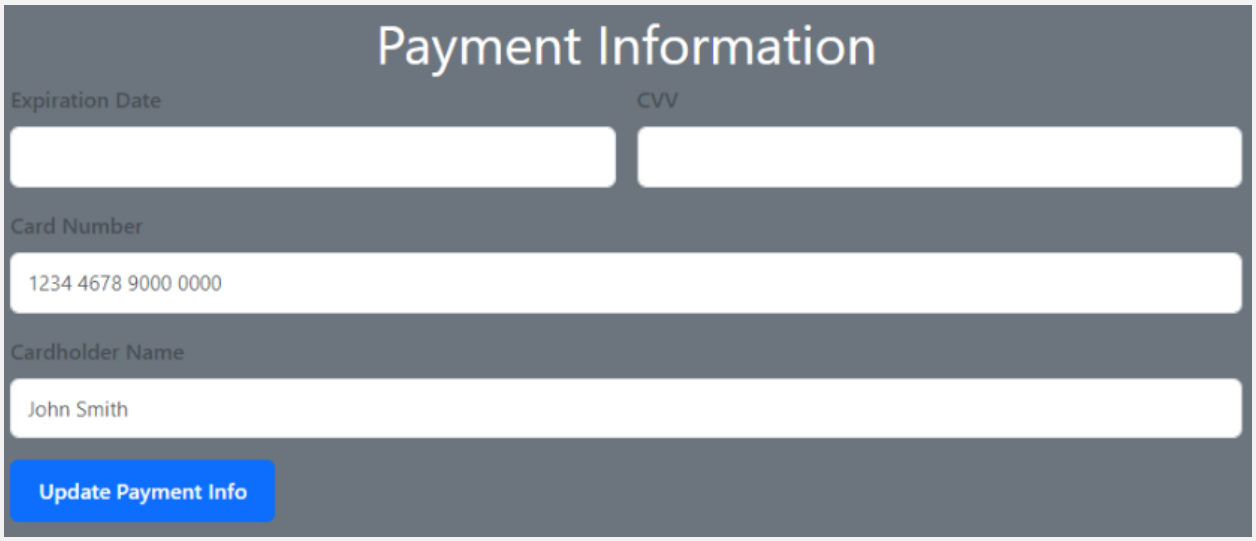


3) Enter your new password in each of the provided fields and then click “Change Password”.



Add Your Payment Information

1) Access your Account Information by clicking your username in the top right corner of the website navigation bar.
2) Enter your payment information into the provided fields and click “Update Payment Info”.

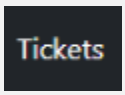


3) After a few seconds, your new payment information will appear at the bottom of the webpage under the “Card Info” section. You may need to refresh the page for it to load.

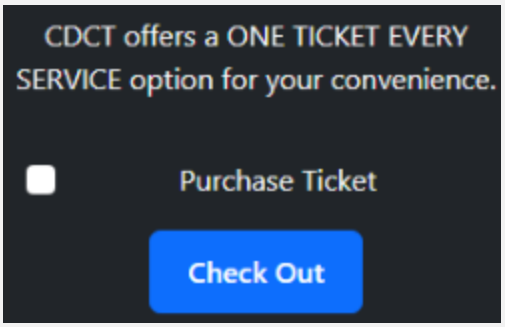
Purchase and Use Transit Tickets

Note: In order to purchase tickets, you must first enter your payment information as described in the above section.

1) In the website navigation bar, click “Tickets”.



2) Our system uses a “one ticket every service” model, meaning a single ticket can be used to access any bus or train route. Check the box next to the “Purchase Ticket” option and click “Check Out”.



3) Your ticket should appear on screen after you click “OK”. Your tickets will remain in your account until to wish to use them. When you are ready to use your ticket, click “Activate Ticket”.

4) You can then scan or delete your tickets after they are active by clicking the desired option.

Transit Administration Website

Overview

The transit administration website is used by Transit Admins to access and update information regarding routes and bus/train details. It is only available to users designated as Transit Admins.

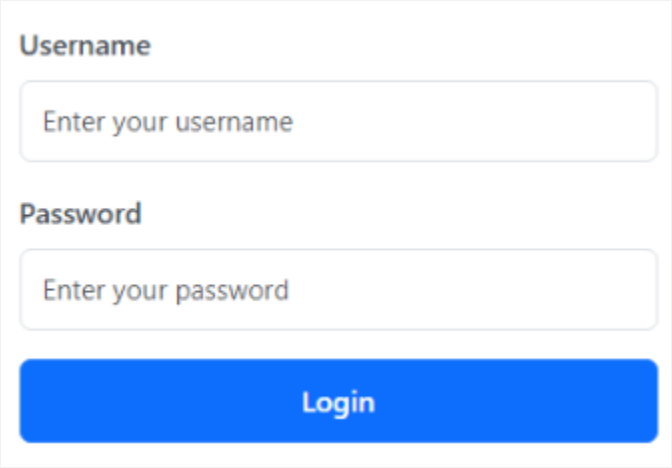
Access the Website

1) Open up a web browser, such as Firefox or Chrome. This can be done by pressing the windows key, typing the browser name, and hitting enter or clicking the application.
2) In the address bar, type admin.team39.isucdc.com and press enter to access the homepage of the website.

Login to Your Account

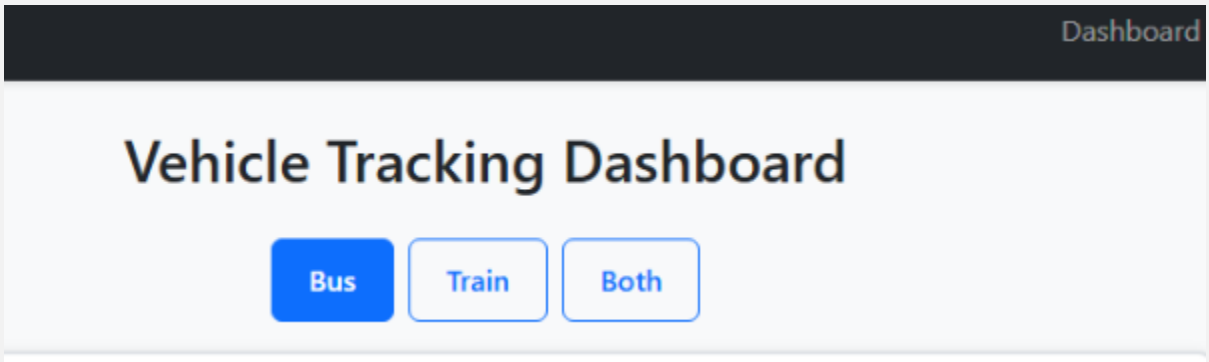
Note: There is a registration feature, however if you are a Transit Admin, you should already have credentials for logging into this website, and newly registered accounts will not have permission to use this website. Please contact your IT administrator if you are a new Transit Admin and have not been provided access to this website.

1) Enter your username and password into the correct fields and click “Login.”



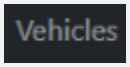
View Bus and Train Routes

1) In the website navigation bar, click “Dashboard”.
2) Select from “Bus”, “Train”, or “Both” to view your desired group of vehicles and their information. Routes and their details will be provided on this page.

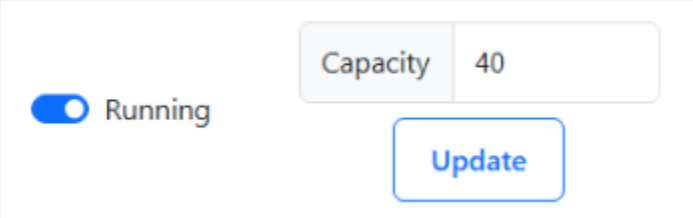


Manage Vehicles in the Fleet

1) In the website navigation bar, click “Vehicles”. Once again, you may filter by buses, trains, or both.



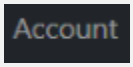
2) Find the vehicle you wish to manage or update. To update a vehicle's capacity, type the desired value into the "Capacity" box. To deactivate or reactivate a vehicle, click the selection bar next to "Running" or "Stopped".



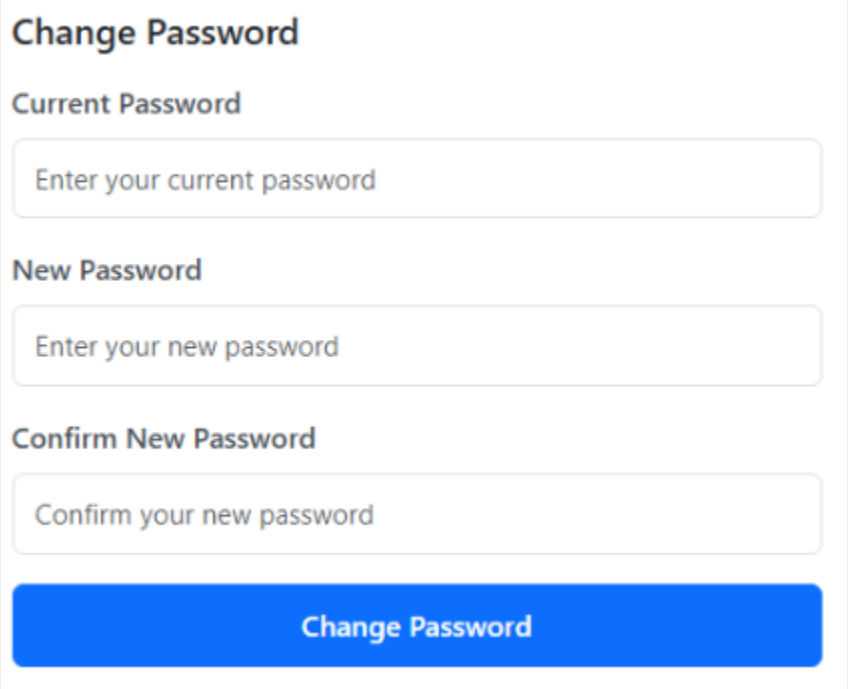
3) Click "Update" to save your changes.

Manage Your Account

1) You may update your Transit Admin account password via the "Account" button on the website navigation bar. Click this button.



2) Enter your current password and your new password into the correct fields and click "Change Password" to change your account password.



3) Click the "Logout" button on this page when you wish to logout of the website.

Remote Administration of WWW and MGMT

Overview

The WWW and MGMT systems are administered remotely via SSH because they are Linux hosts. In addition to SSH access, both systems expose web services for operational use and validation.

Hostnames and Ports

WWW (Linux)

IT Administrators have the necessary permissions to remotely access this machine via SSH.

- SSH: www.team39.isucdc.com:22
- HTTP: www.team39.isucdc.com:80
- Web login path: http://www.team39.isucdc.com/login

MGMT (Linux)

IT Administrators have the necessary permissions to remotely access this machine via SSH.

- SSH: mgmt.team39.isucdc.com:22
- HTTP: mgmt.team39.isucdc.com:5000
- Routes path: http://mgmt.team39.isucdc.com/routes

SSH Access Procedure (Linux)

Use an SSH client (OpenSSH recommended). If you are not sure how to use this or do not have it installed, proceed to the below section titled “Using PuTTY for SSH Access”.

Windows (PowerShell / Windows Terminal)

Press the windows key, type cmd.exe, and press enter. Then, run the following command and enter your credentials. If asked to accept a host fingerprint, enter yes.

- WWW:
    - ssh <username>@www.team39.isucdc.com -p 22
- MGMT:
    - ssh <username>@mgmt.team39.isucdc.com -p 22

macOS / Linux

Open your terminal, run the following command and enter your credentials. If asked to accept a host fingerprint, enter yes.

- WWW:
    - ssh <username>@www.team39.isucdc.com -p 22
- MGMT:
    - ssh <username>@mgmt.team39.isucdc.com -p 22

Post-Login Validation (Recommended)

After connecting via SSH:

- Confirm identity and host:
    - whoami
    - hostname
- Confirm network/service bindings (one of the following):
    - ss -tulpn
    - netstat -tulpn

After connecting via SSH, you will have the ability to remotely administer the machine.

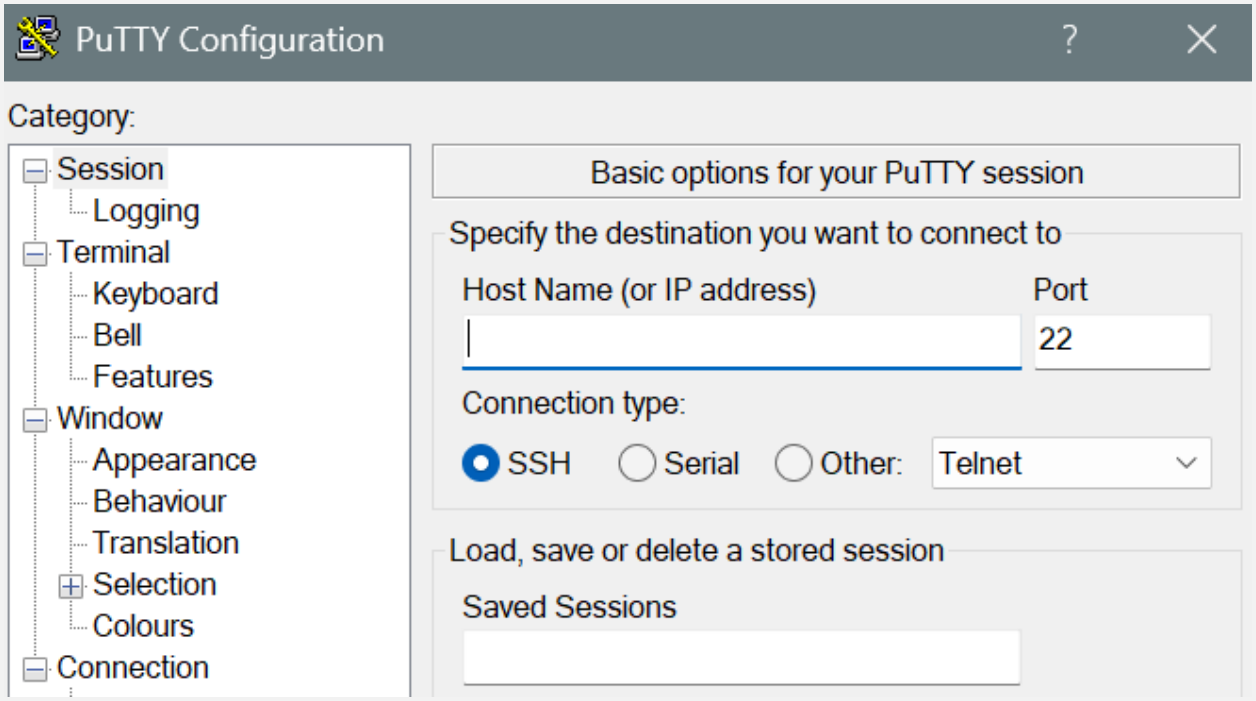
Using PuTTY for SSH Access

Follow the instructions in this section to use your SSH access if you do not have an SSH client configured in your command line.

1) Download and install the correct PuTTY client for your operating system from the following link, if you do not have it already.

https://www.chiark.greenend.org.uk/~sgtatham/putty/latest.html

2) Launch PuTTY by clicking the desktop icon, or by pressing the windows key, typing "putty", and pressing enter. You should see a program similar to the following:



3) Enter the correct hostname as described above. Ensure that Port is set to 22 and Connection Type is set to SSH.
4) Click "Open" to initiate the SSH session. If you are asked to accept a host key, enter yes. You should now be able to administrate the system via SSH.

Remote Administration of ADMIN and TICKETS

Overview

The ADMIN and TICKETS systems are administered remotely via RDP because they are Windows hosts. Both also expose application endpoints over HTTP for user/admin workflows and service health validation.

Hostnames and Ports

ADMIN (Windows)

IT Administrators have the necessary permissions to access this machine via RDP.

- HTTP: admin.team39.isucdc.com:80
- Web login path: http://admin.team39.isucdc.com/login
- RDP: admin.team39.isucdc.com:3389

TICKETS (Windows)

IT Administrators and Ticket Maintenance have the necessary permissions to access this machine via RDP.

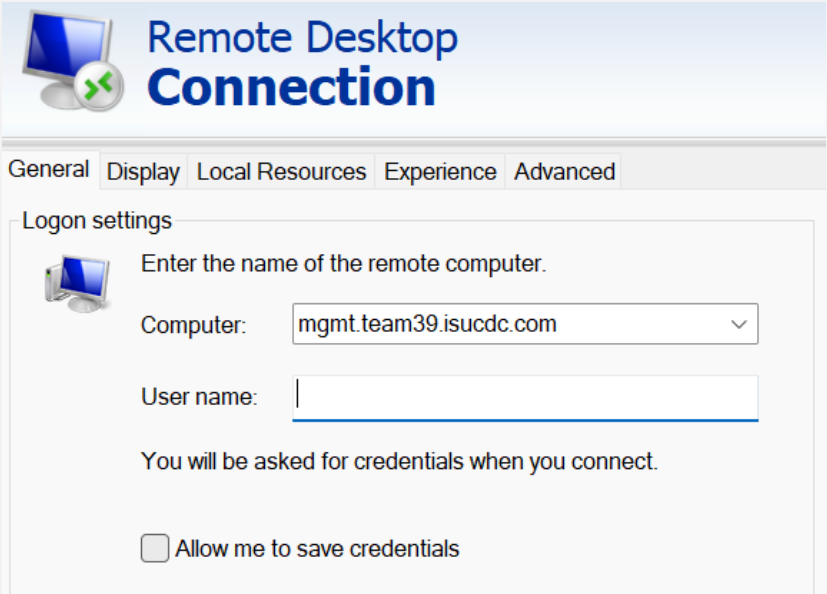
- HTTP health endpoint: tickets.team39.isucdc.com:5000/api/tickets/health
- RDP: tickets.team39.isucdc.com:3389

RDP Access Procedure (Windows)

Use a Remote Desktop client.

Windows (Remote Desktop Connection)

1. Press Win + R, type mstsc, press Enter.
2. In Computer, enter:
    - admin.team39.isucdc.com (ADMIN), or
    - tickets.team39.isucdc.com (TICKETS)
3. Authenticate using the appropriate account format:
    - Domain account: <DOMAIN>\<username>
        - Our domain is TEAM39.
    - Local account: .\<username>



4. After login, confirm the host:

    - Open Command Prompt and run: hostname

macOS (Microsoft Remote Desktop) / Linux (Remmina)

- Host: admin.team39.isucdc.com or tickets.team39.isucdc.com
- Port: 3389
- Username format follows the same domain or local conventions above.

Service Validation (Recommended)

- ADMIN web interface:
    - http://admin.team39.isucdc.com/
    - http://admin.team39.isucdc.com/login
- TICKETS health endpoint:
    - http://tickets.team39.isucdc.com:5000/api/tickets/health
    - curl -i http://tickets.team39.isucdc.com:5000/api/tickets/health

Remote Administration of Domain Controller AD

Overview

The Active Directory Domain Controller is a Windows host. Administrative access is performed via RDP. The directory services are exposed via LDAP/LDAPS for authentication and directory queries used by applications and administrative tools.

Hostname and Ports

AD (Windows Domain Controller)

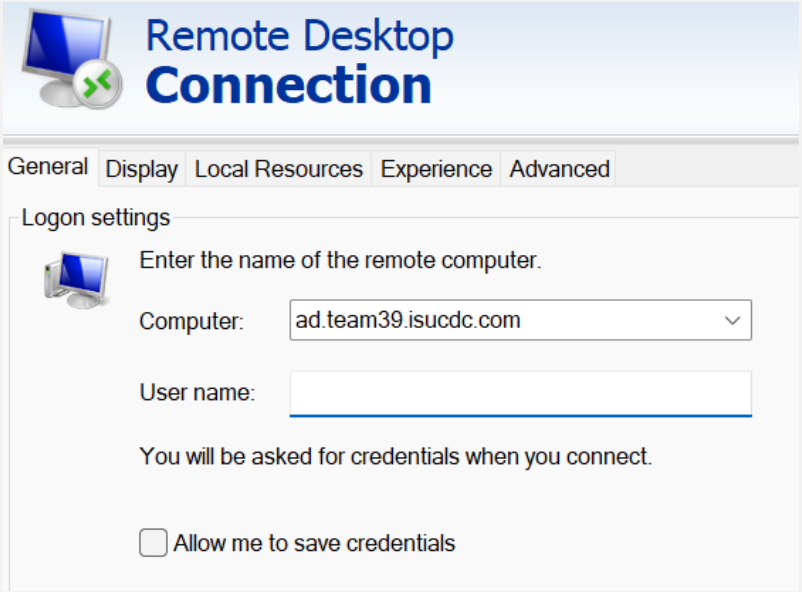
IT Administrators have the necessary permissions to access this machine via RDP.

- LDAP: ad.team39.isucdc.com:389
- LDAPS: ad.team39.isucdc.com:636
- RDP: ad.team39.isucdc.com:3389

RDP Access Procedure (Windows)

1. Open Remote Desktop (mstsc).

2. In Computer, enter: ad.team39.isucdc.com
3. Authenticate using a domain account:
    - <DOMAIN>\<username>
        - Our domain is TEAM39.



4. After login, access the command line by running cmd.exe and confirm the host:
    - Run hostname

LDAP/LDAPS Notes (Service Access)

- LDAP (389) and LDAPS (636) are used by services and tools to communicate with the directory. They are not interactive login methods, however connections to them can be tested to confirm the services are alive.
- To test, or if applications require directory binding, ensure LDAP (389) or LDAPS (636) is reachable by opening a command prompt and running the following:
    - Test-NetConnection ad.team39.isucdc.com -Port 389
    - Test-NetConnection ad.team39.isucdc.com -Port 636